SENTINEL AI

Submission Readiness Report

Where the build stands against the v1.2 Master, Technical and Toolchain & Compliance documents, what is left, and the plan for the final hours before submission.

Competition	AI Builder Cup 2026 · BFSI — Intelligent Risk, Fraud & Financial Experiences
Report date	17 October 2026, evening (IST). Freeze was 15 October; submission is 18 October.
Repository	github.com/Mohammed-Ilyas-AI/Sential-AI · branch `main` · last pushed commit `5d594c3`
Live application	`https://sentinal-ai.cloud.run/` and `/control-tower` (Cloud Run, asia-southeast1)
Supersedes	Phase 1 Completion Report and Phase 1 Audit Corrections Report (both 13 October)
Evidence	Code inspection of `main`; the live URL opened on 17 October; the local `npm run doctor` output; the two earlier reports' test results (285 in `npm run verify`, 26 on Atlas, per commit `97ab06f`)

1. Verdict

Deployed and publicly reachable on Google Cloud — entry requirement met. The UI, sentinel-core, sentinel-stream and the simulator job run on Cloud Run, built by Cloud Build on every push to `main`. The Control Tower loads live data on the public URL; DEMO_TAKEOVER_01 is visible end to end (score 95, 65.4% drain, HOLD under SG-RD-50, system release after the cooling period).

Gemini is not answering on the live application — the most urgent gap. On 17 October the live Investigations page, after pressing *Explain*, showed: "No Gemini model is configured for this environment." The Vertex AI integration is fully built and tested in code, but the deployed service has no model IDs (`_GEMINI_FAST_MODEL` and `_GEMINI_DEEP_MODEL` are empty in `cloudbuild.yaml`, and so is the local `.env`). Until a real Gemini response appears on the live URL, "meaningful use of Google AI" (C§2) cannot be shown, and that criterion sits inside the 40% Technical Merit & GenAI weight (M§4). This is a configuration step of about 20 minutes, not new code (section 6).

4 of 8	3 of 8	1 of 8	1 live URL
MVP modules done (1, 2, 3, 5)	partial (4 intervention, 6 Gemini not live, 7 graph)	not started (8 case report)	on Cloud Run, public

2. Competition entry requirements (M§4, C§2)

REQUIREMENT	STATUS	EVIDENCE AND WHAT REMAINS
Functional prototype, not a concept	Met	Seven deterministic rules, SG-RD-50 gate with hold cascade, live stream, Control Tower, Alerts & Cases with analyst actions, Investigation workspace.
Deployed on Google Cloud	Met	Four Cloud Run workloads in asia-southeast1, Cloud Build trigger on `main`, Atlas reached through a Cloud NAT reserved IP, MongoDB URI in Secret Manager.

Meaningful use of Google AI	Not yet shown	Vertex AI gateway, structured output, schema and sourceField validation, two model tiers, cache and fallback all built and tested with a scripted model. **No real Gemini response on the live URL** — model IDs not configured.
English-only submission material	Met so far	Code, docs and UI are English. Keep it so for the video captions and proposal.
Team of 2–4, working professionals 21+	Owner to confirm	Registration closed 11 October; confirm the team record on the competition site.

3. MVP modules — Master Document §7

MODULE	STATUS	WHAT EXISTS	GAP
1 Transaction simulator	Done	Canonical events (tokenised IDs, eventId, sequence, idempotency key); seeded generators; DEMO_TAKEOVER_01; run-scoped runs so every replay produces fresh activity; Cloud Run Job.	Noise generator; one-button scenario launcher is a stretch item.
2 Real-time risk engine	Done	Seven pure rules ported from the HTML reference; `POST /v1/decision/transactions` answers ALLOW / HOLD / REJECT with versions and trace ID.	—
3 Rapid-drain detection	Done	S$50k / >50% / 24h gate, three exclusions, post-threshold hold cascade (ADR-008).	—
4 Intervention workflow	Mostly	Automatic hold with decision trace; release after customer verification, customer cancellation and fraud confirmation in the UI with two-step confirm; 24h auto-release; append-only audit; cascade as automatic kill-switch behaviour.	"Contact customer" action and an analyst-initiated kill switch button.
5 Investigation workspace	Done	Evidence bundle assembled and displayed (caps 50 tx / 2 hops / 40 nodes), deterministic attribution first, the exact object sent to Gemini, timeline. Live on the public URL.	—
6 Gemini explanation	Code done, not live	Vertex AI REST `generateContent`, responseSchema, versioned system instruction, fast/deep tiers from configuration, sourceField grounding, no-release rule, retry once, deterministic fallback, cache by assessmentId.	Model IDs not set on Cloud Run; no real response observed. Pre-warm the demo account's explanation.

7 Entity graph	Backend only	Network observations ingest, entities and edges with provenance, two-hop flagged-link query feeding ENTITY_LINK scoring.	No graph API for the UI; `neighbourhood` and `traceFunds` not implemented; Network Analysis page is a shell.
8 Case report	Not started	Cases page and component shells only.	One-click report, edit, sign-off, export; CaseService.

4. Surfaces and the ten demo scenes (M§5, M§6)

SURFACE / SCENE	READY	NOTES
5.1 Live Operations Command Center	Yes	Control Tower, live on the public URL.
5.2 Account risk panel	Yes	Alerts & Cases: balance at window start, 24h outflow, drain vs 50%, score, five signals.
5.3 Intervention	Mostly	Decision trace and analyst actions present; Contact customer and manual kill switch missing.
5.4 Investigation workspace	Yes, AI pending	Works; the Gemini panel shows the honest fallback until models are configured.
5.5 Network analysis	No	Shell page.
5.6 Case report	No	Shell page.
Scenes 1–5 (normal ops → intervention)	Yes	Scores 37, 49, 75, 95; drain crosses 50% on the fourth transfer; automatic HOLD; notification.
Scene 6 Investigation opens	Yes	Evidence bundle on screen.
Scene 7 Gemini explains	Blocked	Needs model IDs on Cloud Run (section 6, step 2).
Scene 8 Network expands	No	Module 7 UI.
Scene 9 Analyst acts	Yes	Confirm fraud with analyst identity in Alerts & Cases.
Scene 10 Report generated	No	Module 8.

5. The local error: `npm run doctor`

The only failing preflight check is **"Atlas is refusing this machine's IP address."** Node 24, the `.env` keys, Atlas DNS and all three ports pass. This is not a code defect: MongoDB Atlas only accepts connections from addresses on its Network Access list, and this computer's public IP has changed since it was added (home and mobile connections change IP often).

1. Atlas → **Security** → **Network Access** → **Add IP Address** → **Add Current IP Address**. Add a comment such as "Ilyas laptop 17 Oct".
2. Wait until the entry shows **Active** (about a minute), then run `npm run doctor` and `npm run demo` again.
3. Do **not** add `0.0.0.0/0`. Do not remove `34.21.228.185/32` — that is the Cloud NAT address the deployed services use.

The deployed application is unaffected: it reaches Atlas through the Cloud NAT IP, which is why the live URL keeps working.

6. Final-hours plan, in priority order

The Master Document's own rule (M§9, M§11) still applies: a module that renders but does not respond is worse than no module, and if time runs short the entity-graph screen is dropped before anything else. Building modules 7 and 8 after 15 October breaks the team's own freeze rule; that is a conscious trade-off for the owner, and the order below protects the submission even if it is stopped at any point.

#	WORK	TIME	WHY, AND DONE MEANS
1	Re-allow this machine's IP on Atlas	5 min	Unblocks local runs and recording. Done: `npm run doctor` all green.
2	Turn Gemini on in the live app	20–30 min	Entry requirement and 40% weight. Done: *Explain* and *Summarise case* on the live URL return a grounded answer, not the fallback.
3	Case report, minimal (module 8)	~2 h	Scene 10 and a second visible Gemini use. One click drafts the report from the case summary, timeline and intervention log; the analyst edits, signs off and exports (print to PDF).
4	Contact customer + kill switch buttons (module 4)	~45 min	Completes M§5.3 as written. Append-only intervention records, same two-step confirm.
5	Entity graph screen (module 7) — *drop first*	~2 h	Only if 2–4 are finished and stable. Otherwise keep the honest shell and describe it as designed.
6	Submission package	~1.5 h	README run-through from a clean clone; one-page architecture diagram; Google AI usage section (models, calls, purpose); proposal PDF separating demonstrated from designed; screenshots of all six surfaces; 3-minute captioned video of the ten scenes; offline fallback copy; tag `submission-v1`.

Step 2 in detail — enabling Gemini on Cloud Run

The model identifiers are configuration, never code (CLAUDE.md, T§7). First check which models the project can call. In Cloud Shell, in project `project-f4c013ec-ed17-479b-999`:

```
for m in gemini-2.5-flash gemini-2.5-pro; do echo "== $m"; curl -s -X POST -H "Authorization: Bearer $(g
cloud auth print-access-token)" -H "Content-Type: application/json" "https://aiplatform.googleapis.com/v
```

```
1/projects/project-f4c013ec-ed17-479b-999/locations/global/publishers/google/models/$m:generateContent"
-d '{"contents":[{"role":"user","parts":[{"text":"Reply with the word OK"}]}]}' | head -c 400; echo; don
e
```

If a model returns 404 or "not found", open Vertex AI → Model Garden and use the current Gemini Flash / Pro IDs listed there instead. The deep tier has a 15-second ceiling; if the Pro model is slower than that on the demo account, use the Flash model for both tiers.

Then apply the verified IDs to the running service immediately (no rebuild):

```
gcloud run services update sentinel-core --region asia-southeast1 --update-env-vars VERTEX_LOCATION=glob
al,GEMINI_FAST_MODEL=gemini-2.5-flash,GEMINI_DEEP_MODEL=gemini-2.5-pro
```

Finally make it permanent so the next push does not reset it: set `_GEMINI_FAST_MODEL` and `_GEMINI_DEEP_MODEL` in the Cloud Build trigger's substitutions (or as the defaults in `infra/cloud-build/cloudbuild.yaml`). The service account `sentinel-core` already holds `roles/aiplatform.user` and the Vertex AI API is enabled by the bootstrap script.

7. Submission checklist — Master Document §12

ITEM	STATUS	NOTE
Team registered and locked	Confirm	Closed 11 Oct.
Deployed, publicly reachable URL on Google Cloud	Done	sentinal-ai.cloud.run
Public repository with a README a stranger can follow	Check	Repo public; README run-through from a clean clone not yet re-tested.
One-page architecture diagram	To do	
Proposal document (problem, regulation, solution, architecture, impact)	To do	Reuse Master §1–3, §8, §13.
Three-minute captioned video, ten scenes	To do	Record after Gemini is live.
Screenshots of all six surfaces	4 of 6	Network and Case report are shells.
Google AI usage documented (models, calls, purpose)	To do	Explanation (fast tier), case summary (deep tier), report draft.
Synthetic data provenance stated	Done	README first paragraph; UI footers.
COSMIC and SRF wording checked against M§3	Done so far	Re-check the proposal and video captions.
Demo scenario reproducible from a clean deploy	Done	Run-scoped simulator runs.
Offline fallback video on the presenting laptop	To do	
Multi-tenant and production claims use the M§13 wording	Done so far	"Designed as a multi-tenant platform; demonstrated as a single deployment on synthetic data."

8. Change made today

The deployed core called Vertex AI in `asia-southeast1` only. The current Gemini models are served from the `global` location, so Cloud Build now has a `_VERTEX_LOCATION` substitution (default `global`) that renders into the sentinel-core manifest; the gateway already handled the global endpoint host. Files: `infra/cloud-`

`build/cloudbuild.yaml`, `infra/cloud-build/scripts/render-manifest.sh`, `infra/cloud-run/sentinel-core/service.yaml`, `docs/deployment/cloud-run-deployment.md`. Rendering was checked with and without the variable. Not yet committed or pushed.

9. Risks for the last day

RISK	MITIGATION
Gemini model ID not available in the project	Test with the curl above before changing Cloud Run; fall back to the Flash model for both tiers.
Cold start during judging (min instances 0)	Set `_MIN_INSTANCES` to 1 for the evaluation window (19 Oct – 6 Nov), then back to 0.
New code breaks the demo after the freeze	Every change goes through `npm run verify` and Cloud Build's test steps; tag `submission-v1` on the last good commit before any new feature.
Video recorded before Gemini works	Record only after step 2 is verified on the live URL.
Local Atlas access lost again	Re-add the current IP; the deployed app does not depend on it.

All data shown is synthetic. Bank A, Bank B and Bank C are fictional participating institutions. COSMIC-inspired cross-institutional risk intelligence, demonstrated on synthetic, privacy-safe data. No real bank or COSMIC integration is claimed.